

FREE PDF

AI Engineering Simplified

From Zero → AI Engineer

LLMs · RAG · AGENTS · NLP · DEEP LEARNING · MLOps

100+ hours of learning, simplified into 32 pages



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THE ONLY AI CHEAT SHEET YOU'LL EVER NEED

32 PAGES · 24 CHAPTERS · 2026 EDITION

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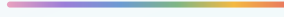
AI Tools Cheat Sheet

Your AI Engineer Roadmap

✦ 23 chapters · 6 parts · Notes by Sandhya Mishra ✦



Part I ~ Foundations



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What is AI?

Definition

Artificial Intelligence (AI) is the science of making machines think and learn like humans. It's about creating computer programs that can understand, reason, learn, and make decisions!

Types of AI

Type	What it means	Example
Narrow AI (ANI)	Good at ONE specific task	Siri, Alexa, Chess engines
General AI (AGI)	Can do ANY intellectual task like a human	Still a dream! Not yet built
Super AI (ASI)	Smarter than all humans combined	Science fiction (for now!)

Key Concepts

Machine Learning — Machines learn from data without being explicitly programmed

Deep Learning — A subset of ML using neural networks with many layers

Natural Language Processing — Machines understanding human language

Computer Vision — Machines understanding images & videos

Robotics — Physical machines that can interact with the real world

Remember: AI is NOT magic — it's math + data + clever algorithms!

Where AI is Used Today

Healthcare — disease diagnosis, drug discovery

Self-driving cars — Tesla, Waymo

Music & entertainment — Spotify recommendations

Your phone — face unlock, autocorrect, photo filters

Shopping — Amazon product recommendations

Chatbots — ChatGPT, Claude, customer service bots



History of AI

1950

Alan Turing proposes the Turing Test — "Can machines think?"

1956

The term "**Artificial Intelligence**" is coined at Dartmouth Conference!

1960s-70s

Early excitement! Rule-based expert systems & ELIZA chatbot

1974-93

AI Winters — funding dried up, progress slowed

1997

IBM's **Deep Blue** beats chess champion Garry Kasparov

2012

Deep Learning revolution! AlexNet wins ImageNet

2016

Google's **AlphaGo** defeats world Go champion Lee Sedol

2017

"**Attention Is All You Need**" paper — Transformer architecture born!

2022-23

ChatGPT & GPT-4 — AI goes mainstream! Everyone is talking about it

2024-26

AI Agents, Multimodal AI, AI Engineers — the age of applied AI

Key Takeaway: AI has had its ups and downs, but we're now in the most exciting era ever!

☆ Essential Math for AI

Don't worry! You don't need to be a math genius — just understand the basics!

Linear Algebra

Vectors — a list of numbers $[1, 2, 3]$ — represents data points

Matrices — a grid of numbers — used in neural network weights

Matrix Multiplication — the core operation in deep learning!

Dot Product — measures similarity between vectors

Statistics & Probability

Mean, Median, Mode — understand your data's center

Standard Deviation — how spread out is your data?

Probability — how likely is an event? $P(\text{rain}) = 0.3$

Bayes' Theorem — update beliefs with new evidence

Normal Distribution — the bell curve! Most data follows this

Calculus (just the basics!)

Derivatives — rate of change — used in gradient descent!

Gradient — direction of steepest increase

Chain Rule — how backpropagation works in neural networks

Partial Derivatives — derivatives for multi-variable functions

Pro Tip: Libraries like NumPy, PyTorch & TensorFlow handle the math for you. Focus on understanding WHAT the math does, not memorizing formulas!

Why Python?

Python is the #1 language for AI because it's simple, readable, and has amazing libraries!

Essential Libraries

Library	What it does	Used for
NumPy	Number crunching & arrays	Math operations
Pandas	Data manipulation	DataFrames, CSV, cleaning
Matplotlib	Plotting & charts	Visualizing data
Scikit-learn	Classic ML algorithms	Classification, regression
PyTorch	Deep learning framework	Neural networks, training
TensorFlow	Deep learning framework	Production ML systems
Hugging Face	Pre-trained models	NLP, LLMs, transformers
LangChain	LLM applications	Chatbots, RAG, agents

Basic Code Pattern

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split

# Load data
data = pd.read_csv("my_data.csv")

# Split into training & testing
X_train, X_test, y_train, y_test = \
    train_test_split(X, y, test_size=0.2)

# Train model
model.fit(X_train, y_train)

# Predict
predictions = model.predict(X_test)
```

Start here: Learn Python basics, then NumPy, Pandas, Scikit-learn, then PyTorch!



Part II ~ Machine Learning



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What is Machine Learning?

Definition

Machine Learning is a subset of AI where machines learn patterns from data to make predictions or decisions — without being explicitly programmed for every rule!

The Three Types of ML

Supervised Learning

Learn from **labeled data**
Like a teacher showing examples!
Spam detection, price prediction

Unsupervised Learning

Find **hidden patterns**
No teacher — discover yourself!
Customer segmentation, anomaly

Reinforcement Learning

Learn by **trial and error**
Get rewards for good actions!
Game AI, self-driving cars

How ML Works (Simple Steps)

- 1 Collect & clean data
- 2 Choose a model/algorithm
- 3 Train the model on data
- 4 Evaluate performance
- 5 Tune & improve
- 6 Deploy to production

Remember: Data quality matters more than model complexity! Garbage in = garbage out



Supervised Learning Algorithms

Classification vs Regression

Classification

Predict a **category**

Is this email spam or not?

What digit is in this image?

Regression

Predict a **number**

What will house price be?

How much rain tomorrow?

Key Algorithms

Linear Regression — Draw a best-fit line through data points. Simple but powerful!

Logistic Regression — Despite the name, it's for classification! Outputs probabilities.

Decision Trees — Like a flowchart of yes/no questions. Easy to understand!

Random Forest — Many decision trees voting together. More accurate!

SVM (Support Vector Machine) — Finds the best boundary between classes.

KNN (K-Nearest Neighbors) — "Tell me your friends, I'll tell you who you are!"

XGBoost / Gradient Boosting — King of tabular data competitions!

Pro Tip: Start with simple models first, then try tree-based models. Don't jump to deep learning for tabular data!



Unsupervised & Reinforcement Learning

Unsupervised Learning Algorithms

K-Means Clustering — Group similar data into K clusters. Like sorting candy by color!

DBSCAN — Finds clusters of any shape. Handles noise well!

PCA — Reduce dimensions while keeping important info. Simplify complex data!

Autoencoders — Neural nets that compress and reconstruct data. Anomaly detection!

Reinforcement Learning (RL)

Agent — the learner/decision maker

Environment — the world the agent interacts with

Action — what the agent does

Reward — feedback (positive or negative)

Policy — the agent's strategy for choosing actions

Famous RL Examples

AlphaGo — Beat the world Go champion

OpenAI Five — Played Dota 2 at pro level

Robot Walking — Robots learning to walk by trial and error

Self-Driving Cars — Learning to navigate roads safely

Note: RL is computationally expensive and needs lots of training. Best for problems where you can simulate the environment!



Model Evaluation & Metrics

Why Evaluate?

A model is only useful if it performs well on NEW, unseen data — not just training data!

Classification Metrics

Metric	What it measures	When to use
Accuracy	% correct predictions	Balanced datasets
Precision	Of predicted +, how many really +?	False positives costly
Recall	Of actual +, how many caught?	False negatives costly
F1 Score	Balance of precision & recall	Imbalanced datasets
AUC-ROC	Overall ranking ability	Binary classification

Regression Metrics

MAE — Average of |actual - predicted|

MSE — Penalizes large errors more

RMSE — Same unit as target variable

R² Score — How much variance explained (1.0 = perfect)

Overfitting vs Underfitting

Overfitting

Model memorizes training data

Great on train, bad on test

Fix: more data, regularization, dropout

Underfitting

Model is too simple

Bad on both train & test

Fix: more features, complex model



Part III ~ Deep Learning



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Neural Networks Basics

What is a Neural Network?

Inspired by the human brain! Layers of connected "neurons" that learn patterns. Each connection has a **weight** adjusted during training.

Architecture

- ~ **Input Layer** — receives your data (features)
- ~ **Hidden Layers** — where the magic happens! Learns patterns
- ~ **Output Layer** — gives the prediction/answer

Key Concepts

- Weights & Biases** — learnable parameters the network adjusts
- Activation Functions** — ReLU, Sigmoid, Tanh — add non-linearity
- Loss Function** — measures how wrong the model is
- Backpropagation** — how the network learns from mistakes
- Gradient Descent** — optimizer that adjusts weights to reduce loss
- Epochs** — number of times model sees entire dataset
- Batch Size** — how many samples processed at once
- Learning Rate** — how big the weight update steps are

Activation Functions

Function	Range	Best for
ReLU	0 to infinity	Hidden layers (default!)
Sigmoid	0 to 1	Binary classification output
Softmax	0 to 1 (sums to 1)	Multi-class output
Tanh	-1 to 1	When you need negatives



CNNs – Computer Vision

What are CNNs?

Convolutional Neural Networks are specialized for processing images. They detect patterns like edges, textures, shapes, and objects!

CNN Building Blocks

Convolutional Layer — Slides filters over image to detect features (edges, corners, textures)

Pooling Layer — Shrinks image while keeping important info. Max Pooling most common.

Flatten Layer — Converts 2D features to 1D for final classification.

Fully Connected Layer — Makes the final prediction based on all features.

Famous CNN Architectures

Model	Year	Key Innovation
LeNet	1998	First practical CNN
AlexNet	2012	Deep learning revolution!
ResNet	2015	Skip connections — 152 layers!
EfficientNet	2019	Smart scaling, efficient
Vision Transformer	2020	Transformers for images!

Applications

Image classification — "Is this a cat or dog?"

Object detection — "Where are the people?"

Face recognition — Face ID on your phone

Medical imaging — detecting diseases in X-rays

RNNs & Sequence Models

What are RNNs?

Recurrent Neural Networks are for sequential data — text, time series, audio. They have memory! Each step remembers info from previous steps.

Types of Sequence Models

Vanilla RNN — Basic version. Has memory but struggles with long sequences

LSTM — Has "gates" to remember or forget info. Solves vanishing gradient!

GRU — Simpler LSTM. Fewer parameters, faster. Often works just as well!

Bidirectional RNN — Reads forward AND backward. Great for context!

Use Cases

Text generation — write poetry, code, stories

Machine translation — English to Hindi

Music generation — compose melodies

Stock price prediction — time series forecasting

Speech recognition — voice to text

Sentiment analysis — positive or negative review?

Note: While RNNs were revolutionary, **Transformers** have largely replaced them because they're faster and handle long sequences better!



Transformers – The Game Changer!

The Big Idea

Transformers use "**Attention**" to process all words in parallel. They are the foundation of GPT, BERT, Claude, and all modern LLMs!

Self-Attention Mechanism

For every word, attention asks: "How important is every other word to understanding *THIS* word?"

Query (Q) — "What am I looking for?"

Key (K) — "What do I contain?"

Value (V) — "What information do I give?"

Transformer Architecture

- 1 **Tokenization** — Break text into tokens
- 2 **Embeddings** — Convert tokens to vectors
- 3 **Positional Encoding** — Add position info
- 4 **Multi-Head Attention** — Multiple perspectives
- 5 **Feed Forward** — Process attended info
- 6 **Stack N layers** — Repeat for depth!

Key Models Built on Transformers

Model	Type	Used For
BERT	Encoder	Understanding text, search
GPT-4	Decoder	Text generation, coding
Claude	Decoder	Analysis, long context
Gemini	Multimodal	Text + images + video



Part IV ~ Language & Generative AI



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NLP – Natural Language Processing

What is NLP?

Natural Language Processing helps machines understand, interpret, and generate human language!

NLP Pipeline

- 1 **Tokenization** — Split text into words/subwords
- 2 **Stopword Removal** — Remove "the", "is", "and"
- 3 **Stemming/Lemmatization** — "running" becomes "run"
- 4 **Vectorization** — Convert text to numbers
- 5 **Modeling** — Train or use pre-trained models

Text Representation

Bag of Words — Count word frequencies. Simple but loses order.

TF-IDF — Weighs words by importance. Rare words score higher.

Word2Vec — Words as vectors. "King - Man + Woman = Queen"

Transformer Embeddings — Context-aware! "Bank" near river is different from "bank" near money

Common NLP Tasks

Sentiment Analysis — is the review positive or negative?

Summarization — condense long text into key points

Named Entity Recognition — find people, places, organizations

Question Answering — answer questions from context

Machine Translation — translate between languages



LLMs – Large Language Models

What are LLMs?

Massive neural networks (billions of parameters) trained on internet-scale text. They generate human-like text, answer questions, write code, and much more!

Major LLMs

Model	Company	Key Strength
GPT-4o / o3	OpenAI	General intelligence, coding
Claude 3.5/4	Anthropic	Long context, safety
Gemini 2.0	Google	Multimodal, huge context
Llama 3	Meta	Open-source, customizable
Mistral	Mistral AI	Efficient, fast
DeepSeek	DeepSeek	Strong reasoning, open

How LLMs are Trained

- 1 **Pre-training** — Learn language from massive text data
- 2 **SFT** — Learn to follow instructions with human examples
- 3 **RLHF** — Align with human preferences via feedback

Key Parameters

Temperature — 0 = deterministic, 1 = creative

Top-p — Controls diversity of outputs

Max Tokens — Maximum output length

Context Window — How much text model can "see"

Parameters — Model size (7B to 1T+)

Tip: You don't always need the biggest model! Smaller fine-tuned models often beat larger general ones for specific tasks.



Prompt Engineering

What is Prompt Engineering?

The art of crafting effective instructions for AI models. A great prompt = a great response!

Prompting Techniques

Zero-Shot — Just ask! No examples needed.

Few-Shot — Give 2-5 examples first. Teaches the pattern!

Chain-of-Thought — "Think step by step" — improves reasoning!

ReAct — Reason + Act. Think, action, observe, repeat.

Role Prompting — "You are an expert..." — sets context

Prompt Formula

```
Role: "You are a senior developer"
Task: "Write a function that..."
Context: "This is for a web API"
Format: "Return as JSON"
Tone: "Explain simply"
Rules: "Do NOT use deprecated methods"
```

Golden Rules:

Be specific and clear

Give examples when possible

Break complex tasks into steps

Specify output format

Iterate and improve your prompts!

☆ Generative AI – Images, Audio & Video

Image Generation

Diffusion Models — Start with noise, denoise to create images. DALL-E 3, Stable Diffusion, Midjourney!

GANs — Generator vs Discriminator battle! One creates, one judges.

VAEs — Encode to latent space, decode to generate new images.

Audio & Speech

Speech-to-Text — Whisper by OpenAI

Text-to-Speech — ElevenLabs, OpenAI TTS

Music Generation — Suno, Udio

Voice Cloning — Clone any voice!

Video Generation

Sora — Stunning videos from text (OpenAI)

Runway Gen-3 — Text/image to video

Pika — Easy video generation

Kling — High-quality long videos

Ethics: Generative AI raises questions about deepfakes, copyright, and misinformation. Use AI responsibly!



Part V ~ Applied AI Engineering



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RAG – Retrieval Augmented Generation

What is RAG?

RAG = search engine + LLM. First retrieve relevant documents, then generate answers based on them!

How RAG Works

- 1 **Chunk** — Split documents into pieces
- 2 **Embed** — Convert chunks to vectors
- 3 **Store** — Save in vector database
- 4 **Query** — User asks a question
- 5 **Retrieve** — Find most similar chunks
- 6 **Generate** — LLM creates answer!

Vector Databases

Database	Type	Best for
Pinecone	Managed	Easy, scalable
ChromaDB	Open source	Local dev
Weaviate	Open source	Hybrid search
Qdrant	Open source	Performance
pgvector	PostgreSQL	Already use Postgres!

Why RAG matters: It lets LLMs answer about YOUR private data without fine-tuning. The #1 pattern in AI engineering today!



Fine-Tuning & AI Agents

Fine-Tuning LLMs

What is Fine-Tuning?

Training a pre-trained LLM further on YOUR data. Like teaching a smart student your domain!

Full Fine-Tuning — Update ALL parameters. Expensive but thorough.

LoRA — Only train small adapters. 10-100x cheaper! Most popular

QLoRA — LoRA + quantization. Fine-tune 70B on one GPU!

AI Agents

Brain (LLM) — Thinks, reasons, plans

Tools — Functions to call (search, code, APIs)

Memory — Short-term + Long-term knowledge

Planning — Break tasks into steps

Feedback Loop — Observe results, adapt

Agent Frameworks

Framework	Best for
LangChain / LangGraph	General agents, complex workflows
CrewAI	Multi-agent teams with roles
AutoGen	Conversational multi-agent
OpenAI Assistants	Quick prototyping



Embeddings & Vector Search

What are Embeddings?

Convert text/images/data into dense number vectors that capture meaning. Similar items = similar vectors!

How Embeddings Work

"King" becomes [0.2, 0.8, -0.1, 0.5, ...] (1536 numbers)

"Queen" becomes [0.3, 0.7, -0.2, 0.6, ...]

"Cat" becomes [-0.5, 0.1, 0.9, -0.3, ...]

King & Queen are close together! Cat is far away.

Popular Embedding Models

Model	Dims	Best for
text-embedding-3-large	3072	High accuracy (OpenAI)
text-embedding-3-small	1536	Good balance (OpenAI)
Cohere embed-v3	1024	Multilingual
BGE / GTE	768	Open source, free!
all-MiniLM-L6	384	Fast & lightweight

Similarity Methods

Cosine Similarity — Angle between vectors. Most common!

Euclidean Distance — Straight-line distance

Dot Product — Quick similarity measure

ANN (HNSW) — Fast search in millions of vectors



Part VI ~ Production & Beyond



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MLOps & Deployment

What is MLOps?

MLOps = DevOps for ML. Deploy, monitor, and maintain ML models in production!

ML Lifecycle

- 1 Data Collection & Processing
- 2 Model Training & Experiments
- 3 Evaluation & Validation
- 4 Deployment (API, edge, batch)
- 5 Monitoring & Drift Detection

Essential MLOps Tools

Category	Tools
Experiment Tracking	MLflow, W&B, Neptune
Model Serving	FastAPI, vLLM, Triton
Containers	Docker, Kubernetes
Data Versioning	DVC, LakeFS
Monitoring	Grafana, Evidently
Cloud	AWS SageMaker, GCP Vertex, Azure ML

Reality: 90% of ML models never make it to production! MLOps bridges the gap.

AI Ethics & Safety

Key Ethical Concerns

Bias & Fairness — AI inherits biases from data. Always audit!

Privacy — Models may memorize personal data. Use anonymization.

Transparency — Users should know when interacting with AI.

Misinformation — LLMs can hallucinate. Deepfakes are real. Verify!

Job Impact — Focus on upskilling & human-AI collaboration.

Safety — Ensuring AI does what we want, not harmful things.

Best Practices

Test for bias before deployment

Document data sources and limitations

Keep humans in the loop for critical decisions

Implement security and access controls

Monitor for model drift and errors

Follow responsible AI frameworks

Remember: With great power comes great responsibility. Let's make AI fair, safe, and beneficial for everyone!

AI Tools Cheat Sheet

Must-Know Python Stack

Tool	Purpose	Priority
Python	Programming language	
NumPy	Numerical computing	
Pandas	Data manipulation	
Scikit-learn	Classical ML	
PyTorch	Deep learning	
Hugging Face	Pre-trained models	
LangChain	LLM applications	
FastAPI	API building	
Docker	Containers	
Streamlit	Quick demos	

Cloud & Compute

Google Colab — Free GPU notebooks. Best for learning!

RunPod / Lambda — Rent GPUs for training

AWS / GCP / Azure — Enterprise cloud platforms

Hugging Face Spaces — Free model hosting

Replicate — Run models via API



Your AI Engineer Roadmap

12-Month Learning Path

Month 1-2

Foundations

Python, NumPy, Pandas, Basic Math, Git

Month 3-4

Classical ML

Scikit-learn, algorithms, EDA, feature engineering

Month 5-6

Deep Learning

PyTorch, CNNs, RNNs, training loop, GPU

Month 7-8

NLP & LLMs

Transformers, HuggingFace, prompt engineering

Month 9-10

Applied AI

RAG, agents, fine-tuning, embeddings

Month 11-12

Production

MLOps, Docker, APIs, monitoring, portfolio

Top Resources

fast.ai — Best free deep learning course

Andrew Ng — ML fundamentals on Coursera

Hugging Face Course — Free NLP course

Kaggle — Practice with competitions

3Blue1Brown — Beautiful math visuals

Andrej Karpathy — DL from scratch

You've got this, Sandhya! Keep learning, keep building, and keep being curious!



Happy Learning!

*You're on your way to becoming an amazing AI Engineer.
Keep building, keep learning, and never stop being curious!*

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